VIRGINIA STANDARDS OF LEARNING

TEST ITEM SET

Algebra I

2009 Mathematics Standards of Learning

Released Spring 2015

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SAMPLE A

What is the solution to $3(2x-1)=3$?

- **A** $x=\frac{1}{3}$
- **B** $x=\frac{2}{3}$
- **C** $x=1$
- **D** $x=5$

Directions: Type your answer in the box. Your answer must be in the form of a fraction in simplest form. Use "/" for the fraction bar.

SAMPLE B

What is the value of $\frac{3}{x+2}$ when $x = 4$?

Your answer must be in the form of a fraction in simplest form.

Which expression represents four less than half a number, n?

- **A** $4-\frac{1}{2}n$
- **B** $\frac{1}{2}n-4$
- **C** $\frac{1}{2}(4-n)$
- **D** $\frac{1}{2}(n-4)$

Which of the following binomials is a factor of $x^2 - x - 6$?

- **A** $x - 1$
- **B** $x - 2$
- **C** $x - 3$
- **D** $x - 6$

Directions: Click on all the correct answers.

Identify each expression that is in simplest radical form.

$x\sqrt{50y}$	$64\sqrt{x}$	$7x^2y\sqrt{2xy}$	$\sqrt{12x^3y^4}$

Which expression is equivalent to $\frac{1}{6}(30x-24y)-\frac{1}{8}(32x-16y)$?

- **A** $x-6y$
- **B** $x-2y$
- **C** $2x-4y$
- **D** $9x-6y$

Which is equivalent to $\sqrt[3]{48}$ in simplest form?

- **A** $2\sqrt[3]{6}$
- **B** $6\sqrt[3]{2}$
- **C** 16
- **D** 24

What is the value of $\sqrt{128}$ in simplest radical form?

- **A** $8\sqrt{2}$
- **B** $64\sqrt{2}$
- **C** $4\sqrt{8}$
- **D** $16\sqrt{8}$

Which polynomial is equivalent to this expression if $n \neq -1$?

$$\frac{3+n-2n^2}{1+n}$$

- **A** $2n-3$
- **B** $3-2n$
- **C** $3-2n^2$
- **D** $4-2n^2$

Which is a factor of $2n^2 - 5n - 42$?

- **A** $2n - 7$
- **B** $2n - 6$
- **C** $n - 7$
- **D** $n - 6$

Which of the following is equivalent to $\frac{a^{12}b^2}{a^3b^6}$?

- A $\frac{a^9}{b^4}$
- B $\frac{b^4}{a^9}$
- C $\frac{a^4}{b^3}$
- D a^9b^4

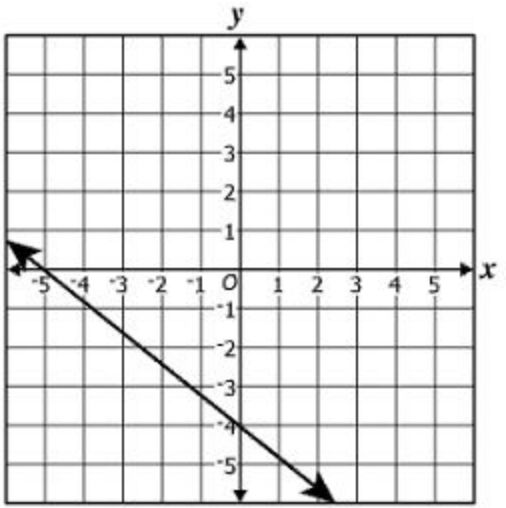
What is the value of this expression when $n = -15$?

$$-2|n + 6|$$

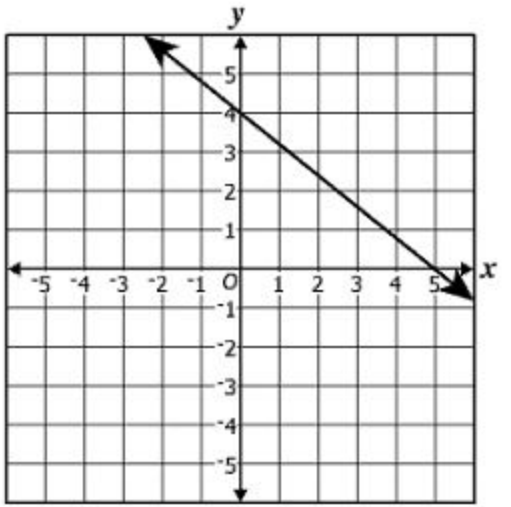
- **A** -42
- **B** -18
- **C** 18
- **D** 42

Which graph best represents the equation $4x + 5y = -20$?

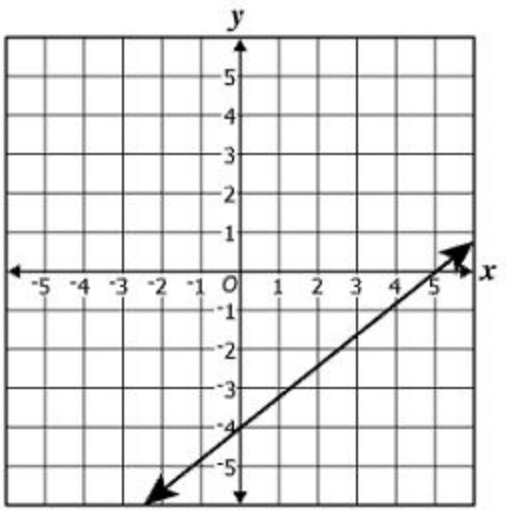
A



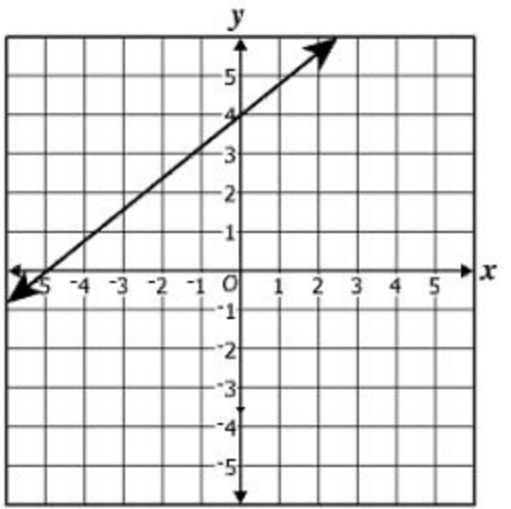
C



B



D



A formula to find the angle measures of an isosceles triangle is shown.

$$180 = 2x + y$$

Which equation can be used to find x ?

- **A** $x = \frac{180 - y}{2}$
- **B** $x = \frac{180 + y}{2}$
- **C** $x = 90 - y$
- **D** $x = 90 + y$

Which equation represents the line that passes through the points $(-4, 4)$ and $(8, -2)$?

- **A** $y = -2x + 14$
- **B** $y = -2x - 4$
- **C** $y = \frac{-1}{2}x + 2$
- **D** $y = \frac{-1}{2}x - 2$

For which system of inequalities is $(^{-}3, 1)$ a solution?

- A $\begin{cases} x + y < -2 \\ 2x - 3y < -9 \end{cases}$
- B $\begin{cases} x + y < -2 \\ 2x - 3y \leq -9 \end{cases}$
- C $\begin{cases} x + y \leq -2 \\ 2x - 3y < -9 \end{cases}$
- D $\begin{cases} x + y \leq -2 \\ 2x - 3y \leq -9 \end{cases}$

What is the solution to this system of equations?

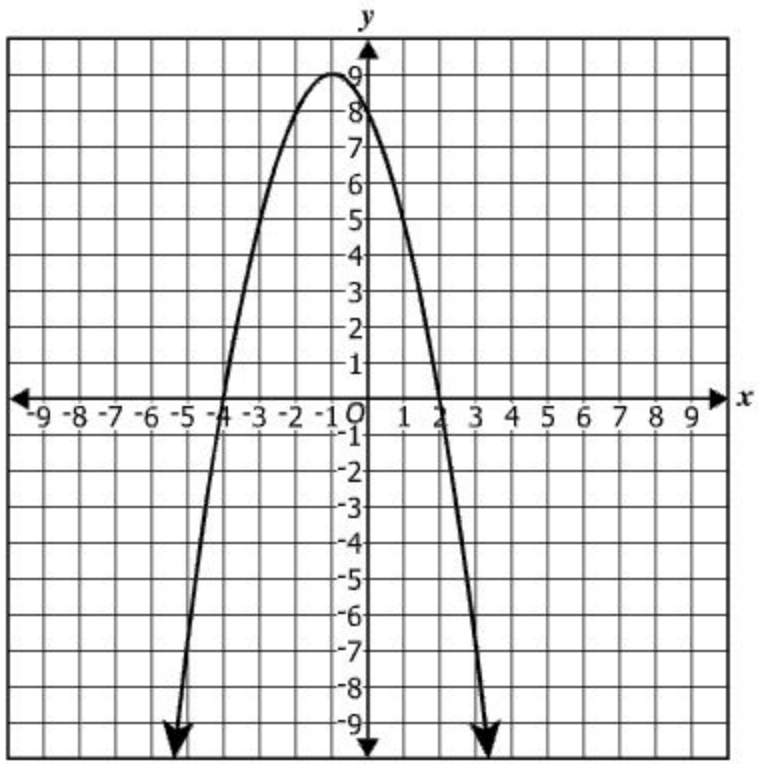
$$\begin{cases} 2x + 4y = 22 \\ 7x + y = 12 \end{cases}$$

- **A** (3, 4)
- **B** (2, $^{-}$2)
- **C** (1, 5)
- **D** ($^{-}$1, 6)

Directions: Click on the grid to plot each of the solutions. You must plot all solutions.

The graph of $y = {}^{-}x^2 - 2x + 8$ is shown.

On the grid, identify each of the solutions to ${}^{-}x^2 - 2x + 8 = 0$.

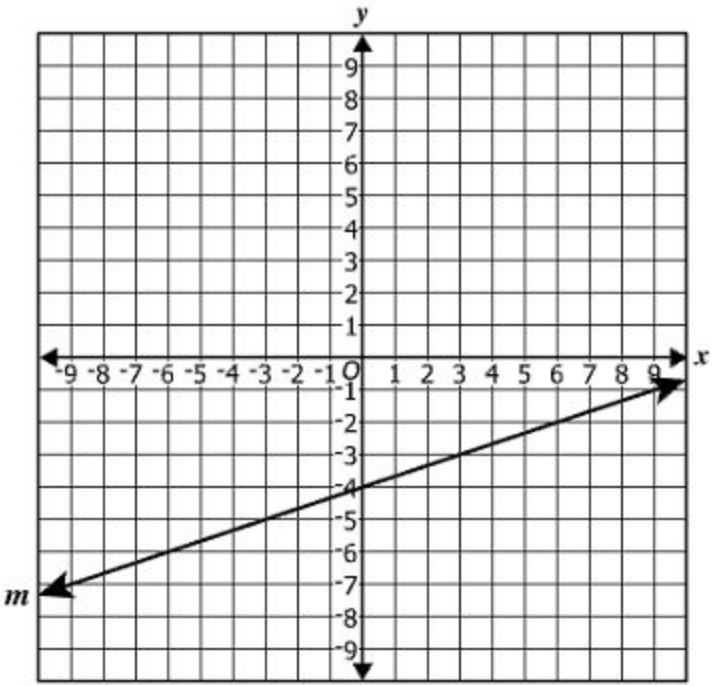


What value of x makes this equation true?

$$3x - 20 = {}^{-}2x$$

- **A** $^{-}20$
- **B** $^{-}4$
- **C** 4
- **D** 20

Which equation best represents line m ?



- **A** $y = -3x - 4$
- **B** $y = \frac{-1}{3}x - 4$
- **C** $y = \frac{1}{3}x - 4$
- **D** $y = 3x - 4$

Directions: Click and drag the answers to the correct boxes.

Christopher incorrectly solved an inequality as shown.

Step 1: $-4(x-7)+1 \leq -3$
Step 2: $-4(x-7) \leq -4$
Step 3: $-4x+28 \leq -4$
Step 4: $-4x \leq -32$
Step 5: $x \leq 8$

Between which two consecutive steps did Christopher make a mistake?

[ ] and [ ]

- Step 1
- Step 2
- Step 3
- Step 4
- Step 5

Directions: Type your answer in the box.

Solve for n:

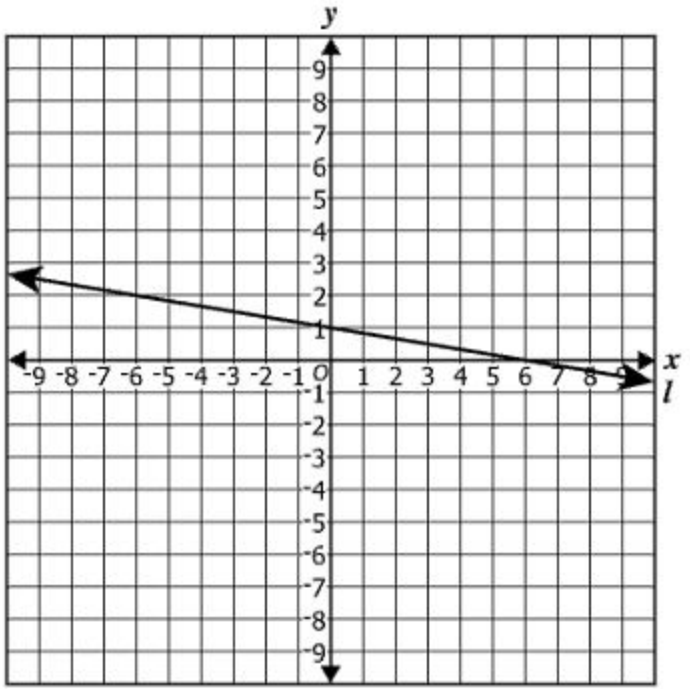
$$\frac{3n-7}{6} = \frac{2n+5}{3}$$

$n =$ ____

What values of x are solutions of $3x^2 + 11x = 20$?

- **A** $\frac{-4}{3}$ and 5
- **B** $\frac{-5}{3}$ and 4
- **C** -4 and $\frac{5}{3}$
- **D** -5 and $\frac{4}{3}$

The graph of line *l* is shown.



Which number is closest in value to the slope of line *l* ?

- **A** -6
- **B** $-\frac{1}{6}$
- **C** $\frac{1}{6}$
- **D** 6

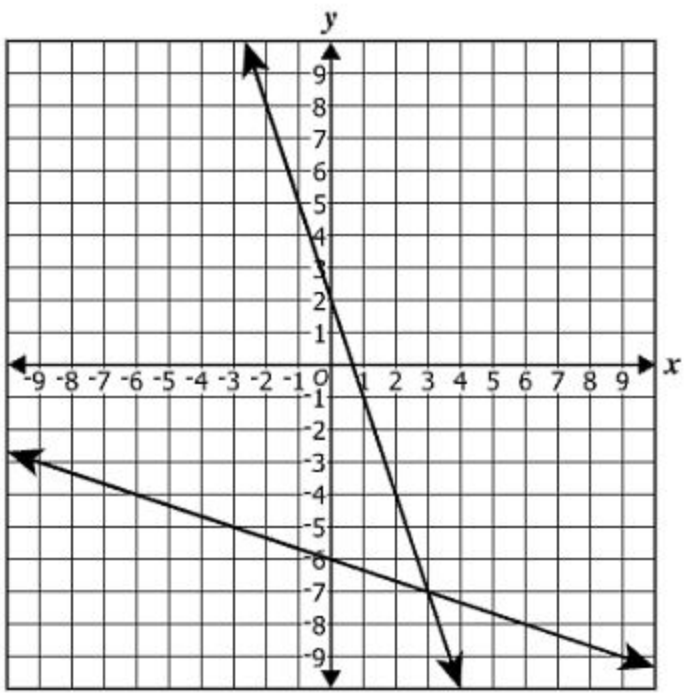
Directions: Type your answer in the box.

Based on the transitive property, complete this statement.

If $2(y-3) \geq 3x-4$ and $3x-4 \geq 6-y$, then $2(y-3) \geq$?

This system of linear equations is graphed as shown.

$$\begin{cases} 3x + y = 2 \\ x + 3y = -18 \end{cases}$$



What is the solution to this system of equations?

- **A** (2, −6)
- **B** (3, −7)
- **C** (−6, 2)
- **D** (−7, 3)

Renee is going bowling.

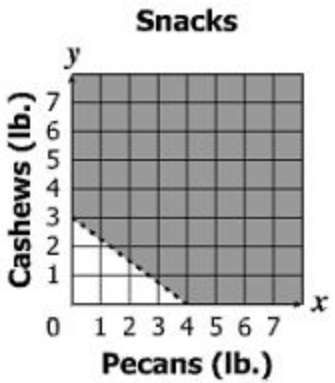
- **The cost per game is $2.50.**
- **Renee will need to rent a pair of bowling shoes for $1.50.**
- **She can spend up to $16.00 to bowl and rent a pair of shoes.**

What is the maximum number of games that Renee can bowl?

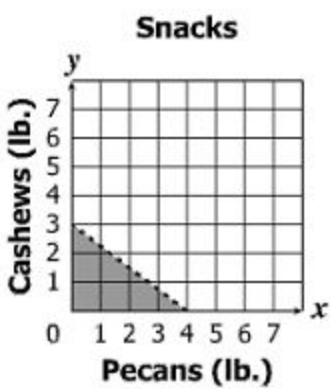
- **A** 4
- **B** 5
- **C** 6
- **D** 9

Malik can spend no more than $24 to buy pecans and cashews. He will pay $6 per pound for pecans and $8 per pound for cashews. Which graph best represents the number of pounds of pecans and the number of pounds of cashews Malik can buy?

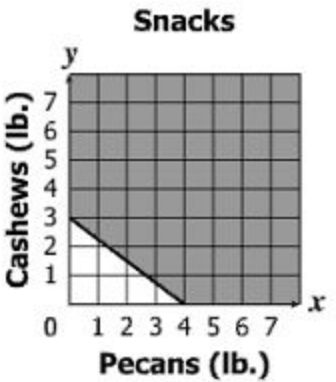
○ **A**



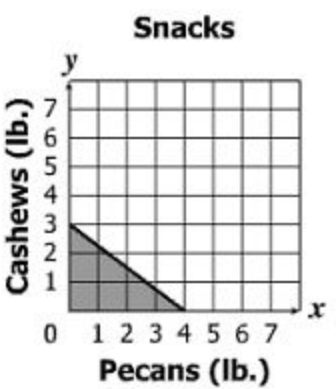
○ **B**



○ **C**

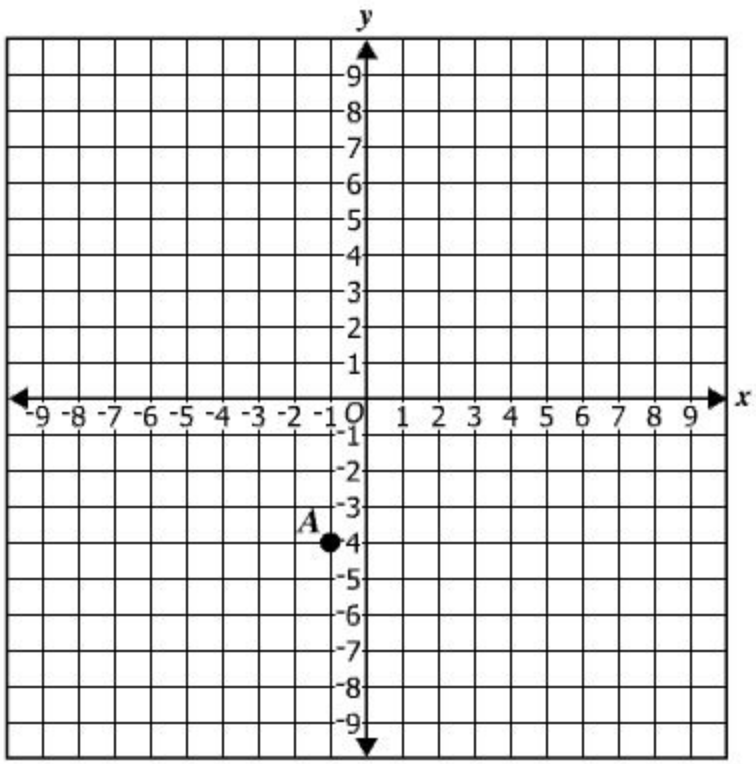


○ **D**



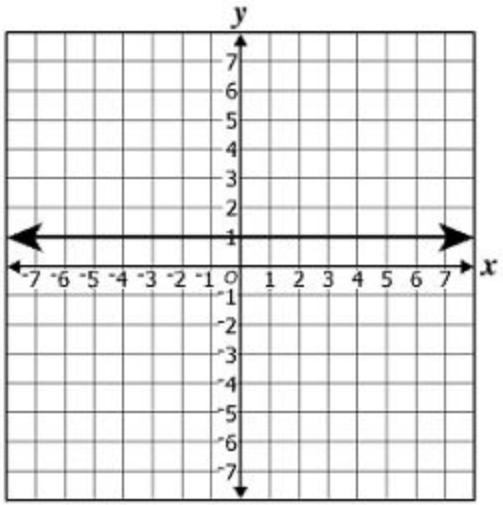
Directions: Click on the grid to plot two points. The coordinates of the points must be integers.

Point A is an element of a direct variation. Plot two points, other than A, that are elements of this direct variation. The coordinates of the points must be integers.

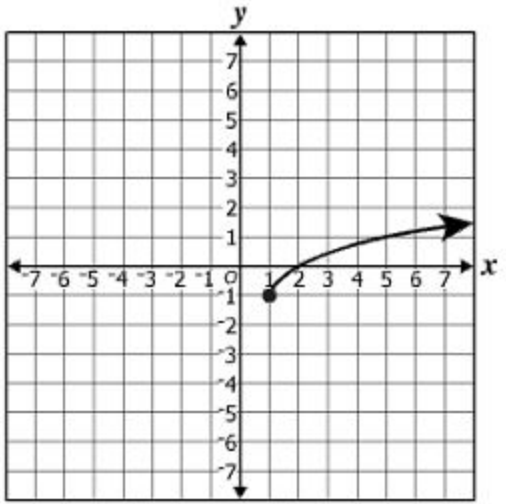


Which graph has exactly one x-intercept and one y-intercept?

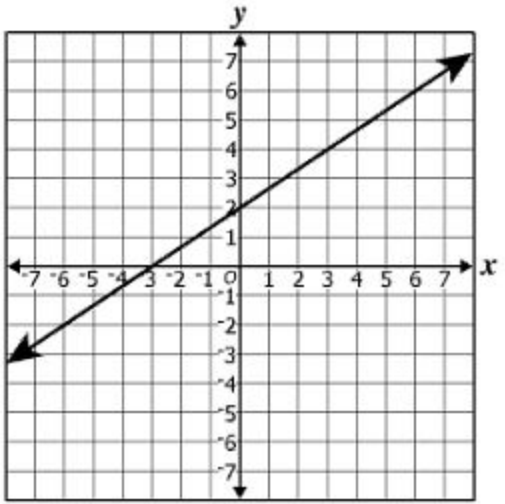
A



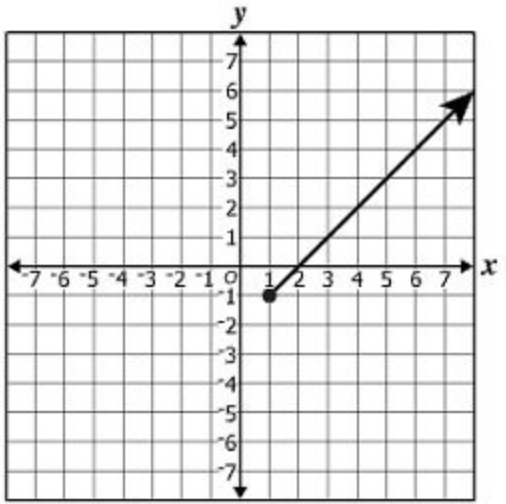
C



B



D



Which equation best represents this data set?

$$\{(-4, -4.8), (-3, -8.2), (-2, -9.1), (-1, -8.1), (0, -4.7), (1, 0.3)\}$$

- **A** $y = 1.1x^2 + 4.2x + 4.9$
- **B** $y = 1.1x^2 + 4.2x - 4.9$
- **C** $y = 1.1x - 4.2$
- **D** $y = 1.1x + 4.2$

A relationship between x and y is shown in this table.

x	y
0	1
1	2
2	5
3	10

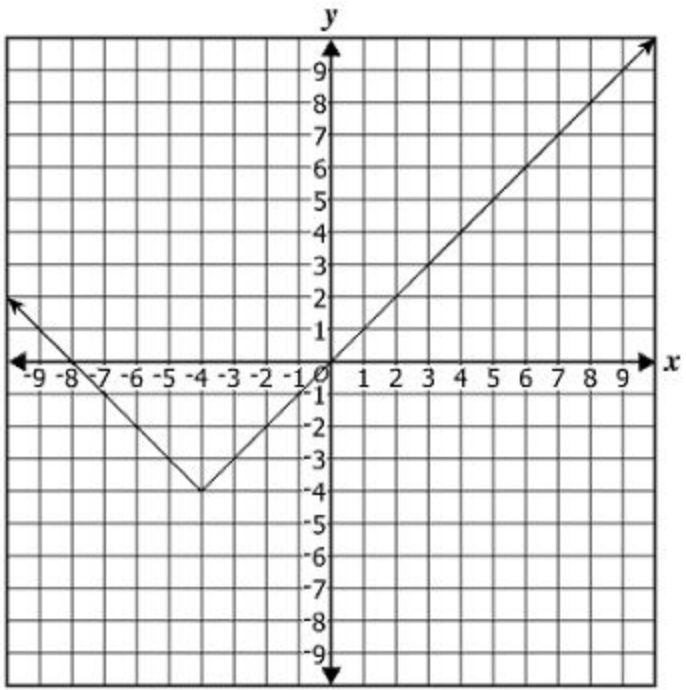
Which equation represents this relationship?

- **A** $y = 2x + 1$
- **B** $y = 5x - 5$
- **C** $y = x^2 + 1$
- **D** $y = (x + 1)^2$

Ms. Scott will pay $2,000 to have her house painted. The amount each painter earns, A, varies inversely for the number of painters, n, that will paint the house. Which equation best represents this situation?

- **A** $A = 2{,}000 + n$
- **B** $2{,}000 = A + n$
- **C** $A = 2{,}000n$
- **D** $2{,}000 = An$

The following graph shows a relation.



Which of the following best describes the range of this relation?

- **A** All real numbers
- **B** All real numbers between -10 and 10
- **C** All real numbers less than or equal to -4
- **D** All real numbers greater than or equal to -4

Directions: Click and drag the answers to the correct boxes.

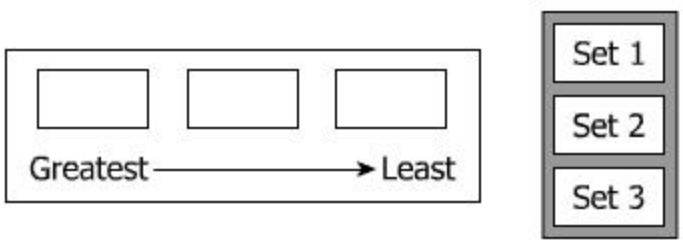
Each of these data sets has a mean of 20.

Set 1: $\{18, 19, 20, 21, 22\}$

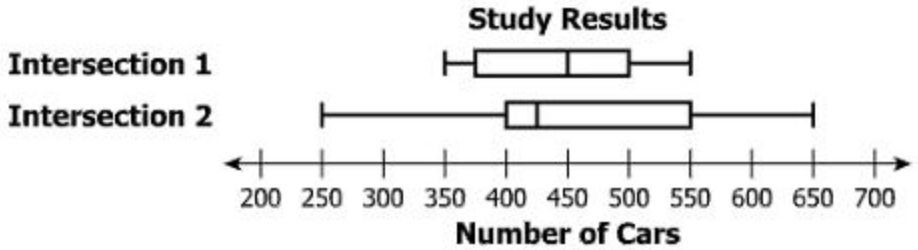
Set 2: $\{20, 20, 20, 20, 20\}$

Set 3: $\{16, 18, 20, 21, 25\}$

Order the sets from greatest standard deviation to least standard deviation.



A study was conducted to determine the number of cars that passed through two intersections each day for 20 days. The results are summarized in these box-and-whisker plots.



Which statement is best supported by these data?

- **A** The range of the data for Intersection 2 is twice the range of the data for Intersection 1.
- **B** The lower quartile for Intersection 1 is greater than the lower quartile for Intersection 2.
- **C** The interquartile range for Intersection 1 is the same as the interquartile range for Intersection 2.
- **D** The total number of vehicles that passed through Intersection 2 is greater than the total number of vehicles that passed through Intersection 1.

Which of these functions has exactly two different zeros?

- **A** $f(x) = \frac{1}{10}x + 4$
- **B** $g(x) = \frac{3x - 10}{3}$
- **C** $h(x) = x^2 - 4x + 4$
- **D** $k(x) = x^2 + 11x + 24$

In which table does y vary directly with x ?

A

x	y
1	3
2	3
3	3

B

x	y
1	4
2	8
3	12

C

x	y
1	5
2	7
3	9

D

x	y
1	9
2	7
3	5

Which equation could represent a graph with x-intercepts of $(4, 0)$ and $(-7, 0)$?

- **A** $y = x^2 + 3x - 28$
- **B** $y = x^2 - 3x - 28$
- **C** $y = x^2 + 3x + 28$
- **D** $y = x^2 - 3x + 28$

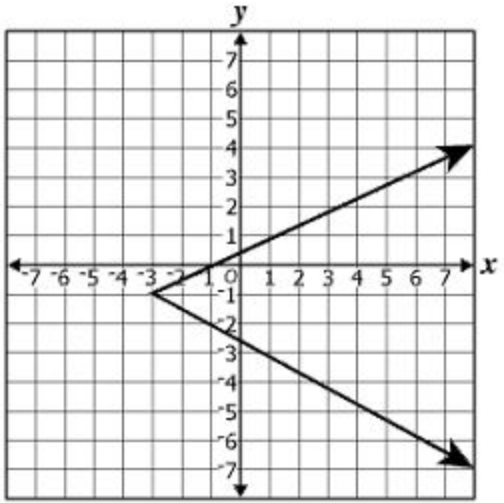
Which number is a zero of the function h ?

$$h(x) = x^2 + 3x - 18$$

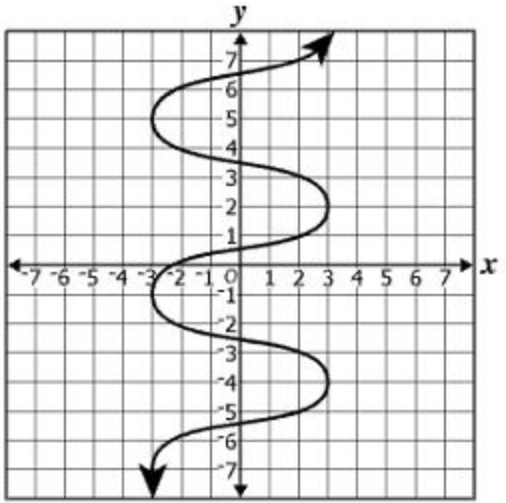
- **A** -6
- **B** -3
- **C** 0
- **D** 6

Which of the following graphs appears to be a function?

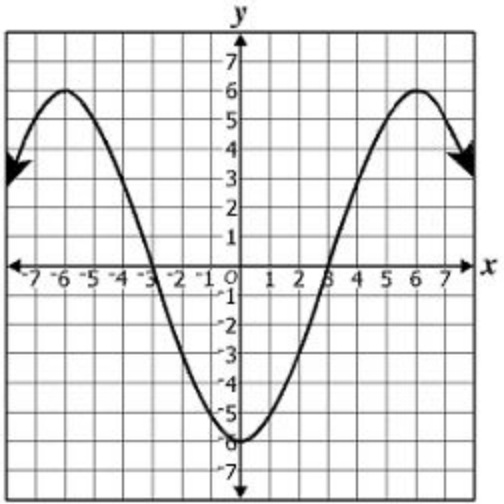
A



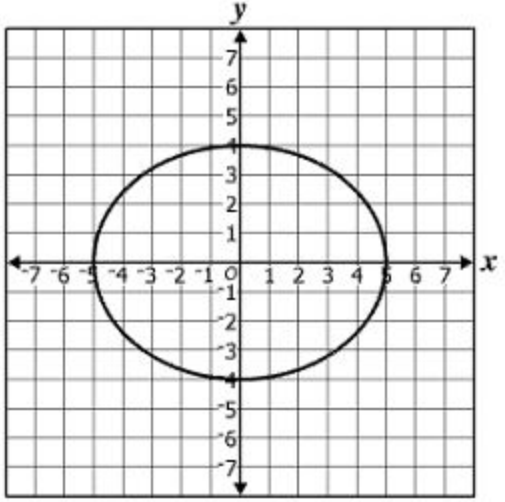
C



B



D



If $f(x) = (x - 3)^2 + 1$, what is $f(6)$?

- **A** -2
- **B** 7
- **C** 10
- **D** 16

Which number is NOT an element in the domain of this relation?

$$\{(-2, 3), (0, 4), (1, 1), (6, 0)\}$$

- **A** 4
- **B** 1
- **C** 0
- **D** -2

$\{(-5, 9), (2, 31), (9, 143), (11, 151), (0, 42), (5, 97)\}$

Using the equation of the line of best fit, which number is the best prediction of the output when the input is 13 ?

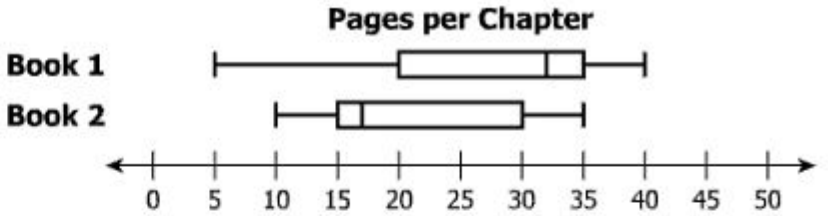
- **A** 127
- **B** 159
- **C** 170
- **D** 178

A data set has a mean of 720 and a standard deviation of 6. Which is closest to the z-score for an element of this data set with a value of 709 ?

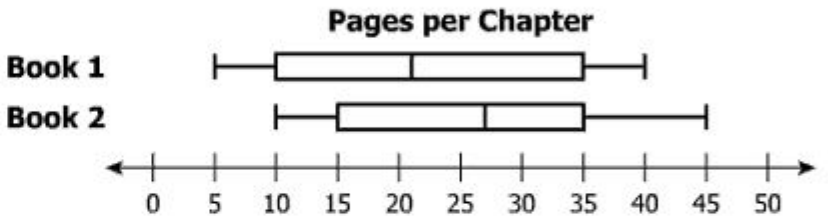
- **A** 11.00
- **B** 1.83
- **C** −11.00
- **D** −1.83

Ramon drew box-and-whisker plots to summarize the number of pages in each chapter of two books. The values of the interquartile ranges for these box-and-whisker plots are the same. Which box-and-whisker plots could represent these data?

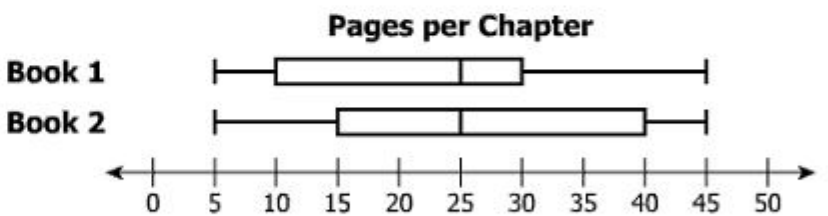
A



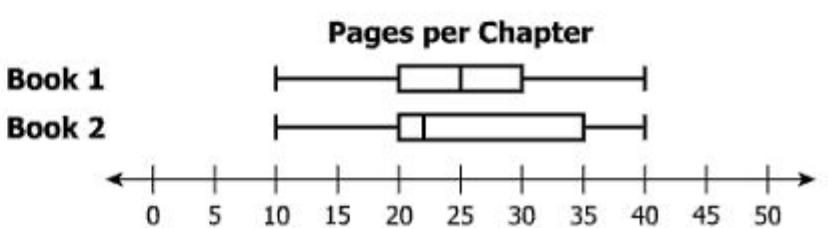
B



C



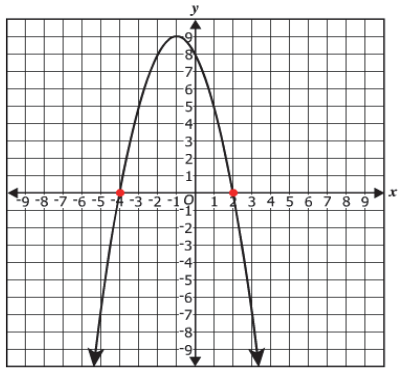
D



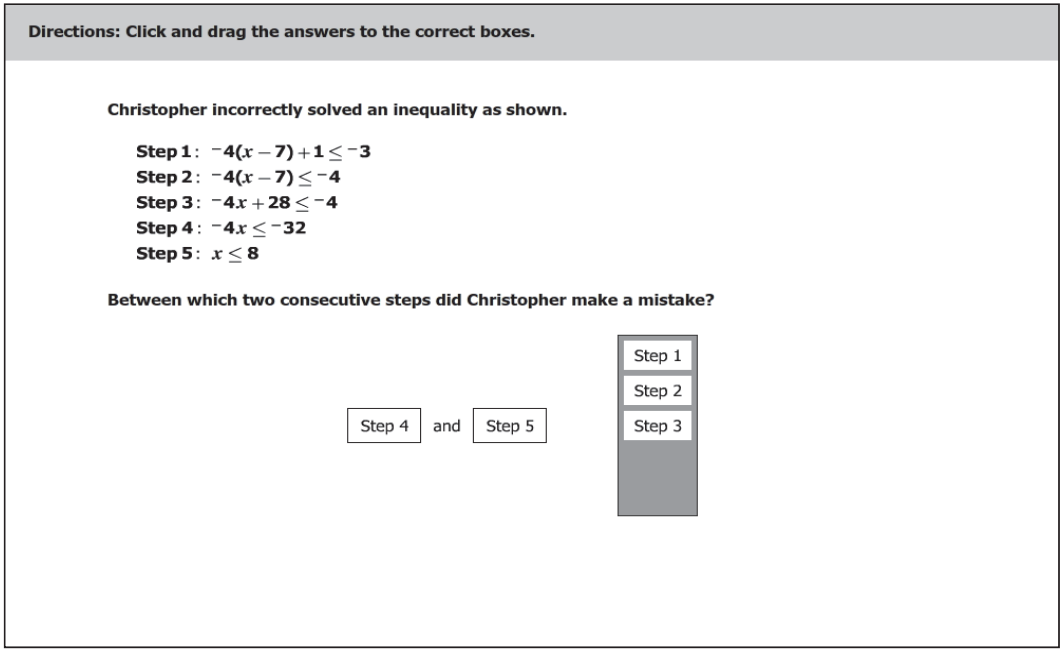
Algebra I
Released Test Item Set Spring 2015
Answer Key

Sequence Number	Item Type: Multiple Choice (MC) or Technology-Enhanced Item (TEI)	Correct Answer	Reporting Category	Reporting Category Description
1	MC	B	001	Expressions and Operations
2	MC	C	001	Expressions and Operations
3	TEI	$64\sqrt{x}$ (second from left) & $7x^2y\sqrt{2xy}$ (third from left) Both of these answers, and only these answers, must be selected. **Directions: Click on all the correct answers.** **Identify each expression that is in simplest radical form.** $x\sqrt{50y}$ $64\sqrt{x}$ $7x^2y\sqrt{2xy}$ $\sqrt{12x^3y^4}$	001	Expressions and Operations
4	MC	B	001	Expressions and Operations
5	MC	A	001	Expressions and Operations

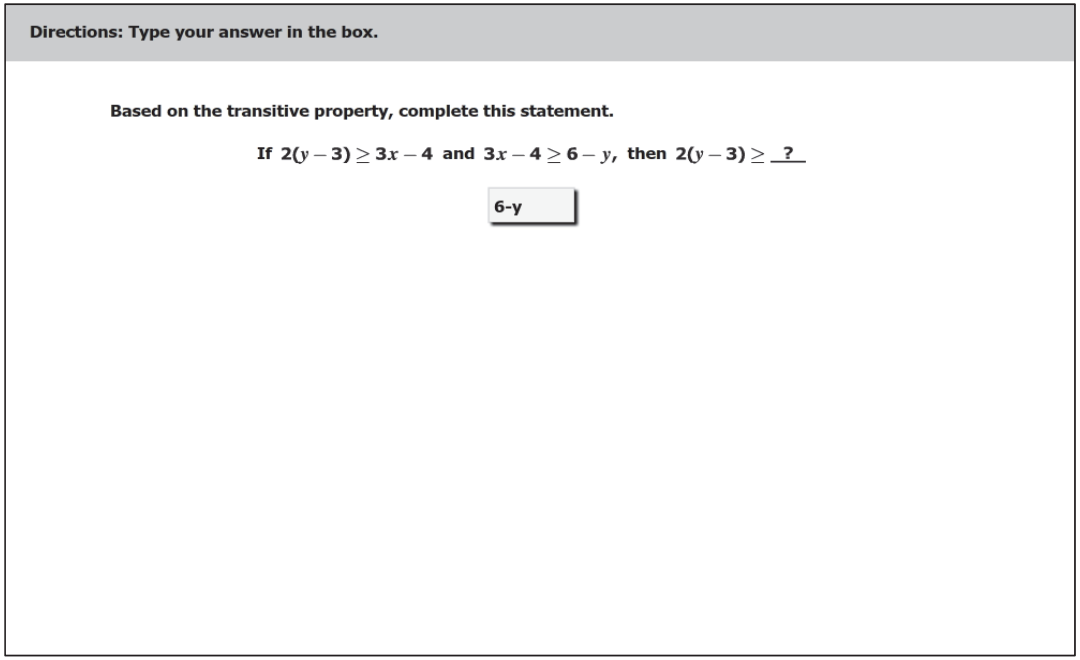
Sequence Number	Item Type: Multiple Choice (MC) or Technology-Enhanced Item (TEI)	Correct Answer	Reporting Category	Reporting Category Description
6	MC	A	001	Expressions and Operations
7	MC	B	001	Expressions and Operations
8	MC	D	001	Expressions and Operations
9	MC	A	001	Expressions and Operations
10	MC	B	001	Expressions and Operations
11	MC	A	002	Equations and Inequalities
12	MC	A	002	Equations and Inequalities
13	MC	C	002	Equations and Inequalities
14	MC	D	002	Equations and Inequalities
15	MC	C	002	Equations and Inequalities

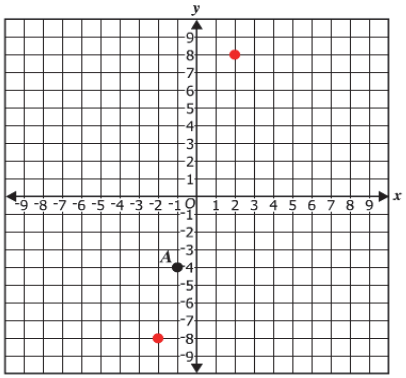
Sequence Number	Item Type: Multiple Choice (MC) or Technology-Enhanced Item (TEI)	Correct Answer	Reporting Category	Reporting Category Description
16	TEI	Both of these points, and only these points, must be plotted on the coordinate plane: (–4,0) and (2,0). **Directions: Click on the grid to plot each of the solutions. You must plot all solutions.** **The graph of $y = -x^2 - 2x + 8$ is shown.** **On the grid, identify each of the solutions to $-x^2 - 2x + 8 = 0$.** 	002	Equations and Inequalities
17	MC	C	002	Equations and Inequalities
18	MC	C	002	Equations and Inequalities

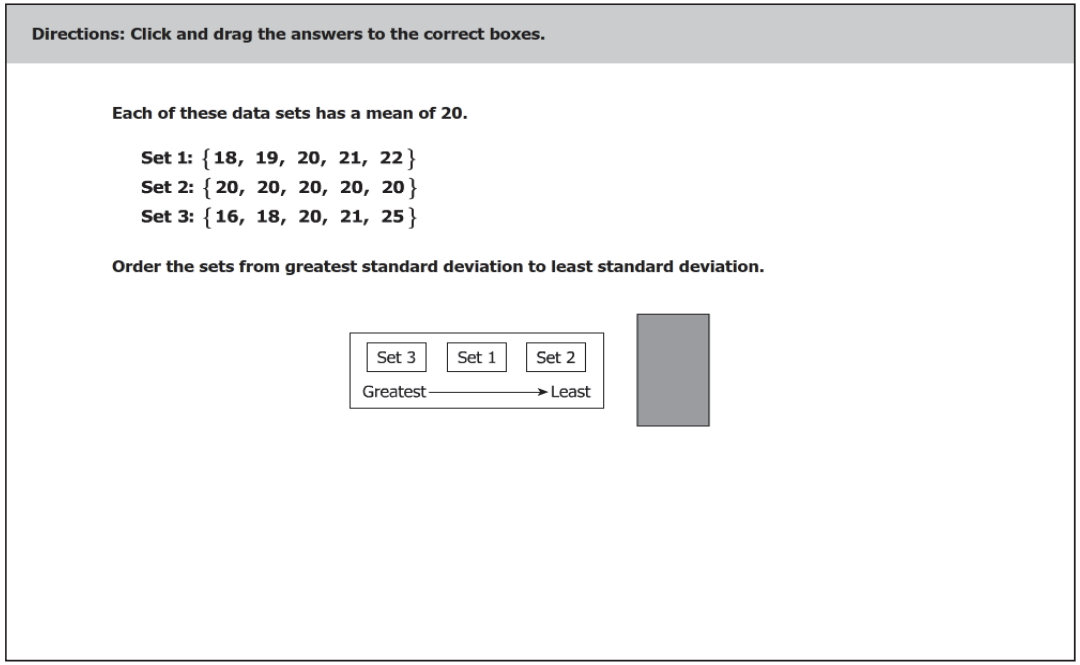
Sequence Number	Item Type: Multiple Choice (MC) or Technology-Enhanced Item (TEI)	Correct Answer	Reporting Category	Reporting Category Description
19	TEI	Step 4 and Step 5 must be placed into the boxes. The order in which they are placed into the boxes does not matter.	002	Equations and Inequalities



Sequence Number	Item Type: Multiple Choice (MC) or Technology-Enhanced Item (TEI)	Correct Answer	Reporting Category	Reporting Category Description
20	TEI	Typed response: -17 (and all equivalent answers) **Directions: Type your answer in the box.** **Solve for n:** $\frac{3n-7}{6} = \frac{2n+5}{3}$ $n =$ -17	002	Equations and Inequalities
21	MC	D	002	Equations and Inequalities
22	MC	B	002	Equations and Inequalities

Sequence Number	Item Type: Multiple Choice (MC) or Technology-Enhanced Item (TEI)	Correct Answer	Reporting Category	Reporting Category Description
23	TEI	Typed response: 6-y OR any equivalent expression that does not exceed six characters 	002	Equations and Inequalities
24	MC	B	002	Equations and Inequalities
25	MC	B	002	Equations and Inequalities
26	MC	D	002	Equations and Inequalities

Sequence Number	Item Type: Multiple Choice (MC) or Technology-Enhanced Item (TEI)	Correct Answer	Reporting Category	Reporting Category Description
27	TEI	Any TWO of these points must be plotted on the coordinate plane: (–2,–8), (0,0), (1,4), or (2,8) Two of these points, (2,8) and (−2,−8), are shown on the coordinate plane below. **Directions: Click on the grid to plot two points. The coordinates of the points must be integers.** **Point A is an element of a direct variation. Plot two points, other than A, that are elements of this direct variation. The coordinates of the points must be integers.**  	003	Functions and Statistics
28	MC	B	003	Functions and Statistics
29	MC	B	003	Functions and Statistics
30	MC	C	003	Functions and Statistics
31	MC	D	003	Functions and Statistics
32	MC	D	003	Functions and Statistics

Sequence Number	Item Type: Multiple Choice (MC) or Technology-Enhanced Item (TEI)	Correct Answer	Reporting Category	Reporting Category Description
33	TEI	Answers must be placed in the correct order from left to right: Set 3; Set 1; Set 2 	003	Functions and Statistics
34	MC	A	003	Functions and Statistics
35	MC	D	003	Functions and Statistics
36	MC	B	003	Functions and Statistics
37	MC	A	003	Functions and Statistics
38	MC	A	003	Functions and Statistics
39	MC	B	003	Functions and Statistics
40	MC	C	003	Functions and Statistics

Sequence Number	Item Type: Multiple Choice (MC) or Technology-Enhanced Item (TEI)	Correct Answer	Reporting Category	Reporting Category Description
41	MC	A	003	Functions and Statistics
42	MC	C	003	Functions and Statistics
43	MC	D	003	Functions and Statistics
44	MC	A	003	Functions and Statistics

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